

Avishkar Pawar

Database Engineer

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SUMMARY

Computer Engineering student with hands-on experience building database internals from scratch. Designed MiniDB with ACID transactions, MVCC, B+ Tree indexing, buffer pool management, SQL-like query processing, replication, and sharding. Also built AI vector database infrastructure with HNSW search, WAL persistence, and FastAPI endpoints. Completed Software Developer internships at OSMOS and Core-Decimal Solutions.

WORK EXPERIENCE

Software Developer Intern, OSMOS

Jul 2025 - May 2026

- Developed backend services using Node.js and Python for application and product development.
- Contributed to product development from scratch by designing, implementing, testing, and improving core features.

Software Developer Intern, Core-Decimal Solutions

Feb 2025 - Aug 2025

- Built full-stack features using Vue.js, Express.js, Sequelize, and MySQL in an MVC architecture.
- Developed reusable Vue.js components, integrated RESTful APIs with Axios, and designed Sequelize data mappings.

Software Development Intern, Campus Credential

2024

- Worked on web application features, database-backed workflows, and production-oriented bug fixes.

PROJECTS

MiniDB - Distributed Key-Value Database | C++17, Boost.Asio, Protocol Buffers, CMake

[GitHub](#)

- Built B+ Tree indexing, MVCC transactions, Raft consensus, SQL-like query processing, sharding, TCP networking, and HTTP monitoring for 10,000+ operations/sec with sub-millisecond local latency.

VectorFlow - AI Vector Database | Python, FastAPI, NumPy, Numba, LangChain

[GitHub](#)

- Built HNSW nearest-neighbor search, WAL persistence, distributed sharding, REST APIs, optimized distance computation, and RAG pipeline integration for semantic search.

StreamFlow - Kafka-like Event Streaming Platform | Java 17, Spring Boot, Netty, Docker

[GitHub](#)

- Implemented partitions, leader-follower replication, ISR tracking, consumer groups, log-based storage, zero-copy transfer, admin APIs, and monitoring hooks for high-throughput messaging.

City-wide Dark Store Network Projection | JavaScript, location analytics

[GitHub](#)

- Built a location analytics tool for projecting dark-store placement across cities using demand, service-area, and accessibility considerations.

EDUCATION

B.Tech. in Computer Engineering, MIT Academy of Engineering, Pune

2022 - 2026

CGPA: 9.24/10

ACHIEVEMENTS

- GATE 2026 Qualified: AIR 3124, GATE Score 601, 51.45/100 marks in Computer Science.
- GATE 2025 Qualified: AIR 9860, GATE Score 453, 39.32/100 marks.
- First Place, Technodium 25 technical competition.
- Certifications: AWS Cloud Practitioner, CCNA, Python Essentials, PCAP, NPTEL Modern C++, NPTEL Machine Learning, NPTEL DSA, NPTEL Cloud Computing, NPTEL Cryptography, Saylor Cryptography, Red Hat Academy.

SKILLS

Focus Areas	Database Internals, ACID Transactions, MVCC, B+ Trees, Query Processing, Vector Search
Languages	C++, Java, Python, JavaScript, TypeScript, SQL, PHP
Frameworks	Spring Boot, FastAPI, React, Node.js, Express.js, Vue.js, Flask
Databases	PostgreSQL, MySQL, MongoDB, Redis, custom database implementations
AI/ML	TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba, computer vision
Tools	Git, Docker, Linux, AWS, CMake, Maven, Prometheus, REST APIs
Core CS	Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design